Date: September 6, 2026
Selected Agent: AI Optical Agent

Problem in the Context of Agent:

In the traditional optical and eyewear industry, manual checking of optical prescriptions, measuring Interpupillary Distance (IPD), and fitting frame dimensions manually takes a lot of time and effort. There is also a high chance of human error during manual data entry and lens fitting calculations, leading to incorrect lens powers and uncomfortable vision correction for patients. To solve this problem, an intelligent **AI Optical Agent** can be developed that automatically scans optical prescriptions, measures facial and pupil dimensions, verifies lens compatibility, assigns precise prescription parameters, and generates an optimized optical recommendation. The agent must ensure 100% accurate optical calculations within minimal time.

Problem Formulation:

a) Textual

- **States:** Raw prescription image/scan, extracted Sphere (SPH), Cylinder (CYL), Axis, Addition (ADD), Interpupillary Distance (IPD), frame suitability dimensions, total parameters verified, status of processing (processed / unprocessed).

- **Initial State:** Patient has submitted the optical prescription image or face scan. All parameters are unextracted/unverified, status is unprocessed, and recommendations are empty.

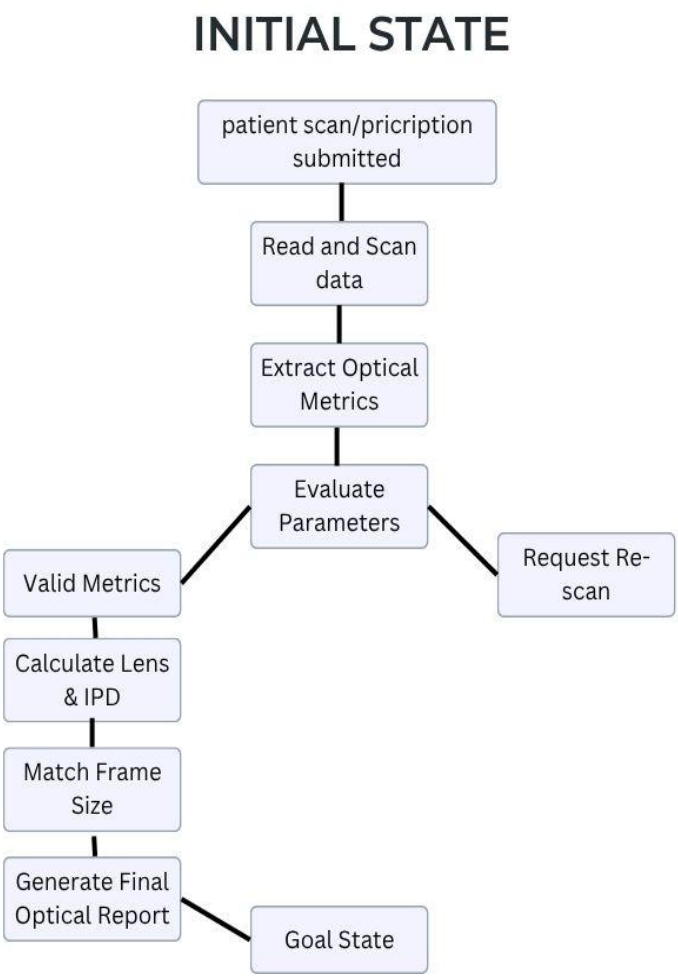
- **Actions:** Capture/read image, perform OCR optical extraction, measure IPD and facial coordinates, compare extracted metrics with standard optical bounds, calculate lens power compatibility, match frame sizing, display final optical prescription report.

- **Transition Model:** When the agent processes one parameter (e.g., SPH or IPD), that parameter becomes "verified". If valid, the values are added to the optical profile; if invalid or unreadable, a re-scan request is triggered and the profile state updates.

- **Goal Test:** All optical parameters (SPH, CYL, Axis, IPD) are fully checked and validated, appropriate frame size is matched, total lens calculations are tabulated, and the final optical report is generated.

- **Path Cost:** Time taken to process the scan + accuracy of optical parameter extraction. The goal is to complete processing with maximum accuracy in minimum time.

a) Graphical



PEAS Evaluation:

- **Performance Measure:** Optical extraction accuracy (100%), IPD measurement precision, processing speed, zero human entry error, customer recommendation satisfaction.

- **Environment:** Optical prescription documents, digital camera/optical scanner input, optical parameter database, frame/lens catalog.

- **Actuators:** Screen display (showing optical breakdown & 3D frame fit), digital prescription output generator, inventory database updater.

- **Sensors:** High-resolution camera / optical document scanner, OCR feature extractor, facial geometry/pupil detection algorithm, touch/keyboard input.

Environmental Properties:

- **Observability:** Fully Observable (the agent has access to all input images, optical rules, and catalog data; nothing is hidden).
- **Deterministic vs Stochastic:** Deterministic (every optical calculation and extraction rule leads to a clear, predictable outcome).
- **Episodic vs Sequential:** Episodic (processing one patient's optical prescription does not affect the processing of another paper or patient).
- **Static vs Dynamic:** Static (the environment and prescription data remain static while being processed by the agent).
- **Discrete vs Continuous:** Discrete (states are clear and distinct: scanned/unscanned, valid/invalid, matched/unmatched).
- **Single vs Multi-Agent:** Single Agent (only the AI Optical Agent is evaluating and generating the prescription report).